

Markscheme

Mathematics: analysis and approaches
Practice question bank

13. (a)

Consider the function $f(x) = \frac{23 - 4x}{(x + 1)(x - 2)^2}$.

Write $f(x)$ as the sum of partial fractions in the form $\frac{A}{x + 1} + \frac{B}{x - 2} + \frac{C}{(x - 2)^2}$.

$$23 - 4x = A(x - 2)^2 + B(x + 1)(x - 2) + C(x + 1) \quad (M1)$$

$$\text{substituting } x = -1: 27 = 9A \Rightarrow A = 3 \quad \text{A1}$$

$$\text{substituting } x = 2: 15 = 3C \Rightarrow C = 5 \quad \text{A1}$$

$$\text{comparing } x^2 \text{ coefficients: } 0 = A + B \Rightarrow B = -3 \quad \text{A1}$$

Note: Accept equivalent valid methods for finding B , e.g. comparing constant terms or substituting a convenient value of x such as $x = 0$.

[4 marks]